Market Size Analysis on Electric Vehicles Industries

Python Project Report

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Dataset: Electric_Vehicle_Population_Data_Excel_sheet.xlsx

Analysis basis: supplied 9-page project brief and supplied EV population Excel file.

Scope note: This dataset contains 177,866 vehicle records and focuses heavily on Washington State. It is a registration/population snapshot, not a complete measurement of all EV sales or market revenue.

1. Executive Summary

The supplied workbook contains **177,866** records and **17** fields. After trimming text, coercing numeric fields and checking exact duplicates, **177,866** records remain; **0** exact duplicate rows were removed. The most represented state is **WA**. The largest county by record count is **King** (92,740 records), and the largest city is **Seattle** (29,447). The leading make is **TESLA** (79,659 records), while the leading model is **MODEL Y** (35,989).

2. Project Objectives

The project assesses model-year patterns, geographic distribution, make/model popularity, EV technology types, electric range and MSRP characteristics, coordinate-based hotspots, and a simple trend projection. The supplied brief also asks for market growth and future forecasting; limitations of the available snapshot are documented below.

3. Dataset and Data Preparation

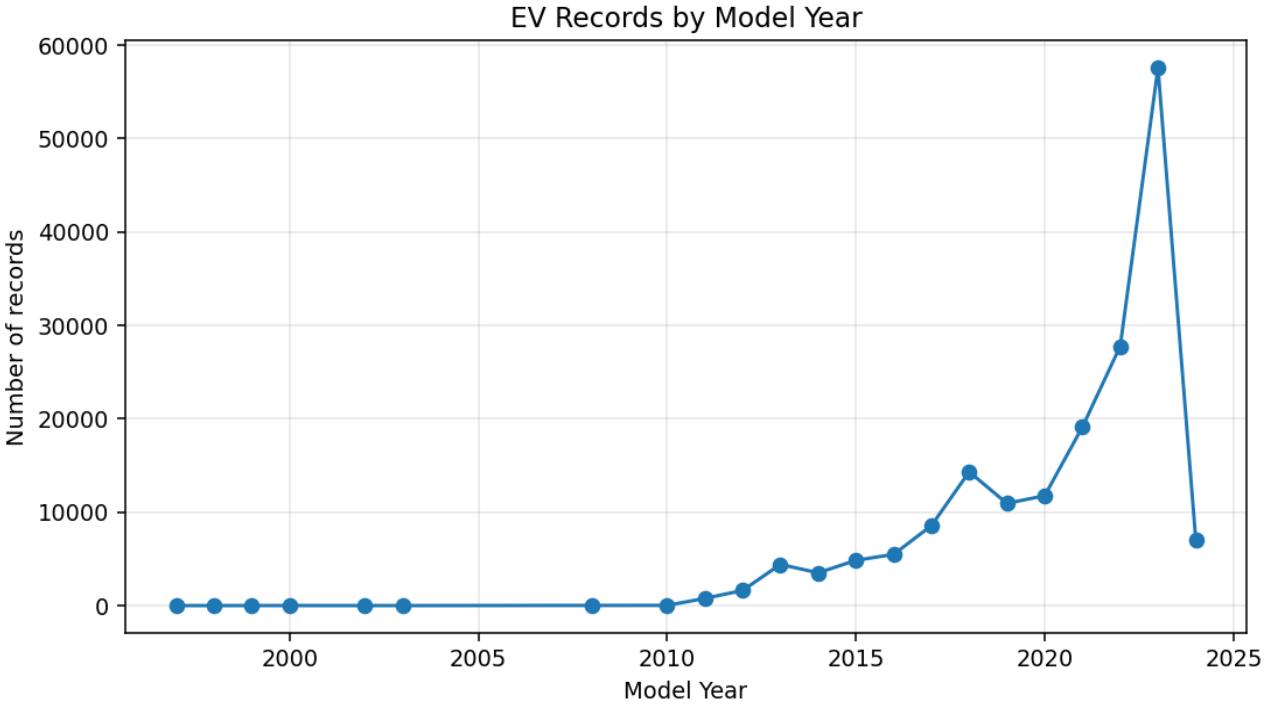
The workbook has 17 columns: VIN (1-10), County, City, State, Postal Code, Model Year, Make, Model, Electric Vehicle Type, Clean Alternative Fuel Vehicle (CAFV) Eligibility, Electric Range, Base MSRP, Legislative District, DOL Vehicle ID, Vehicle Location, Electric Utility, and 2020 Census Tract. Text fields were trimmed, numeric fields coerced to numeric types, and exact duplicates checked. Missing location values: **{summary['missing_location']}**; missing electric-range values: **{summary['missing_range']}**; missing MSRP values: **{summary['missing_msrp']}**. Zero range and MSRP values were retained in the cleaned dataset because they may indicate unavailable or unreported values; positive values were used for the range/MSRP descriptive statistics.

4. Key Findings

• **Geography:** WA is the most represented state. King County has 92,740 records and Seattle has 29,447. These are raw record counts, not per-capita adoption rates.
• **Manufacturer:** TESLA is the most common make with 79,659 records.
• **Model:** MODEL Y is the most common model with 35,989 records.
• **Range:** among positive Electric Range values, mean range is 121.8 miles and median is 84.0 miles.
• **MSRP:** among positive Base MSRP values, mean is $57,078 and median is $59,900. MSRP may be zero or stale for some vehicles and is not a transaction price.
• **Range and MSRP:** Pearson correlation among rows with both fields positive is 0.406, indicating a moderate positive association in this subset, not causation.

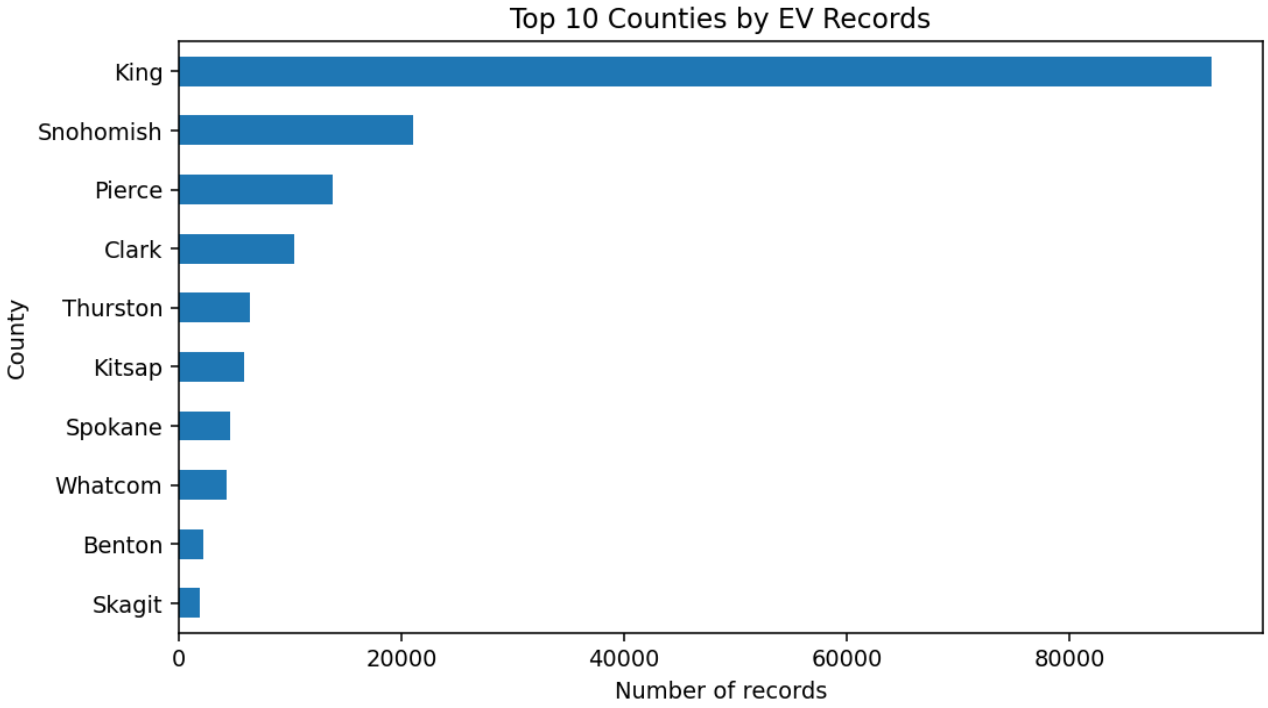
5. Visual Exploration

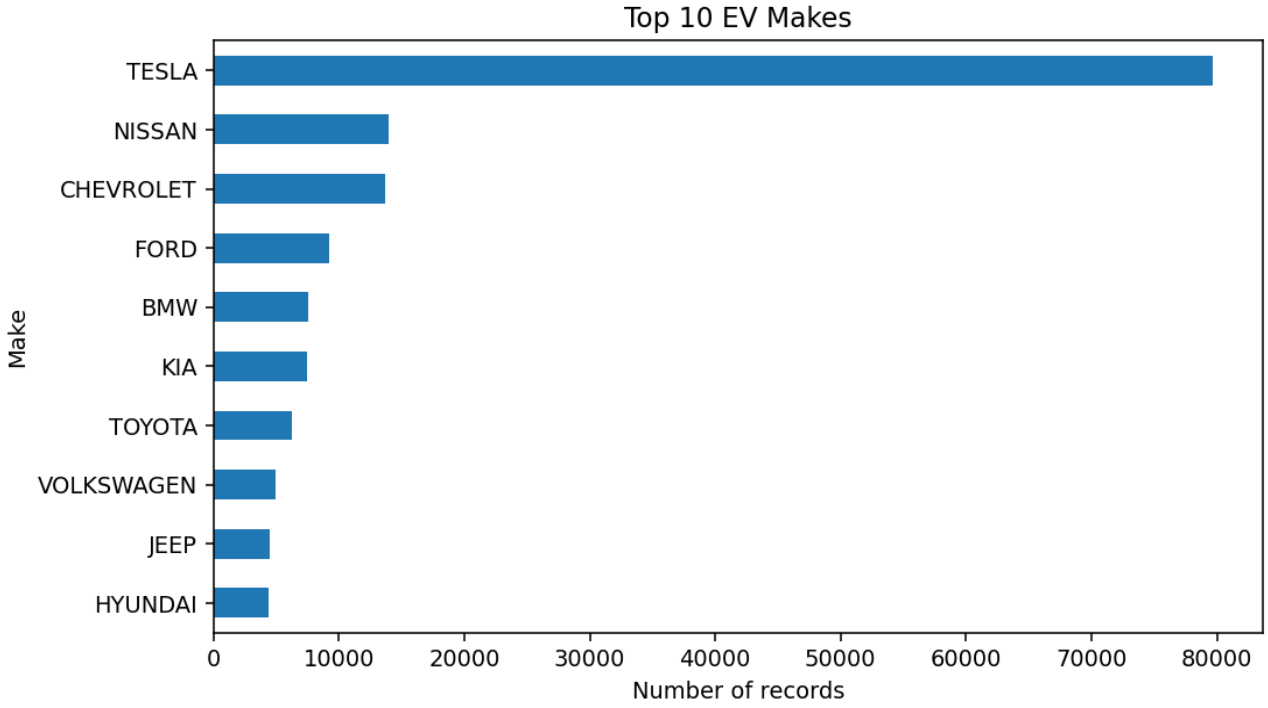
EV records by model year



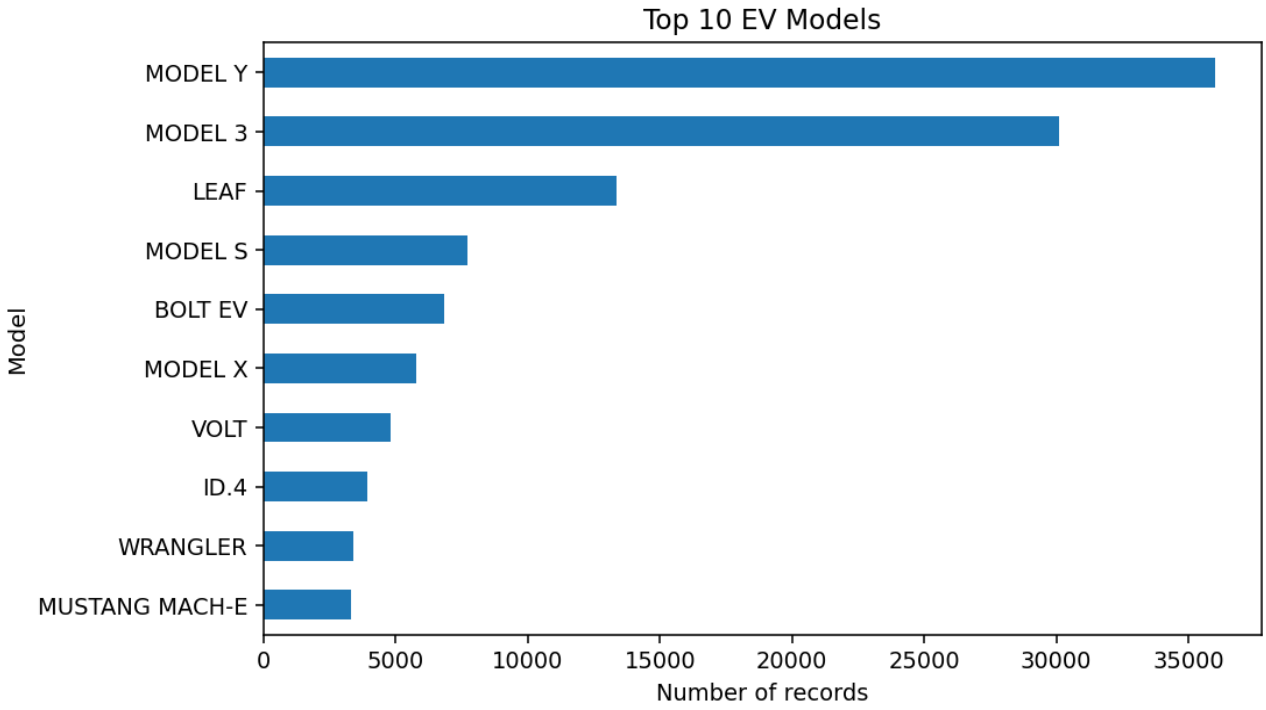
The highest count by model year in this file is model year **2023**, with **57,587** records. Model year is not the same as registration date; therefore, this chart describes the model-year composition of the snapshot, not annual sales.

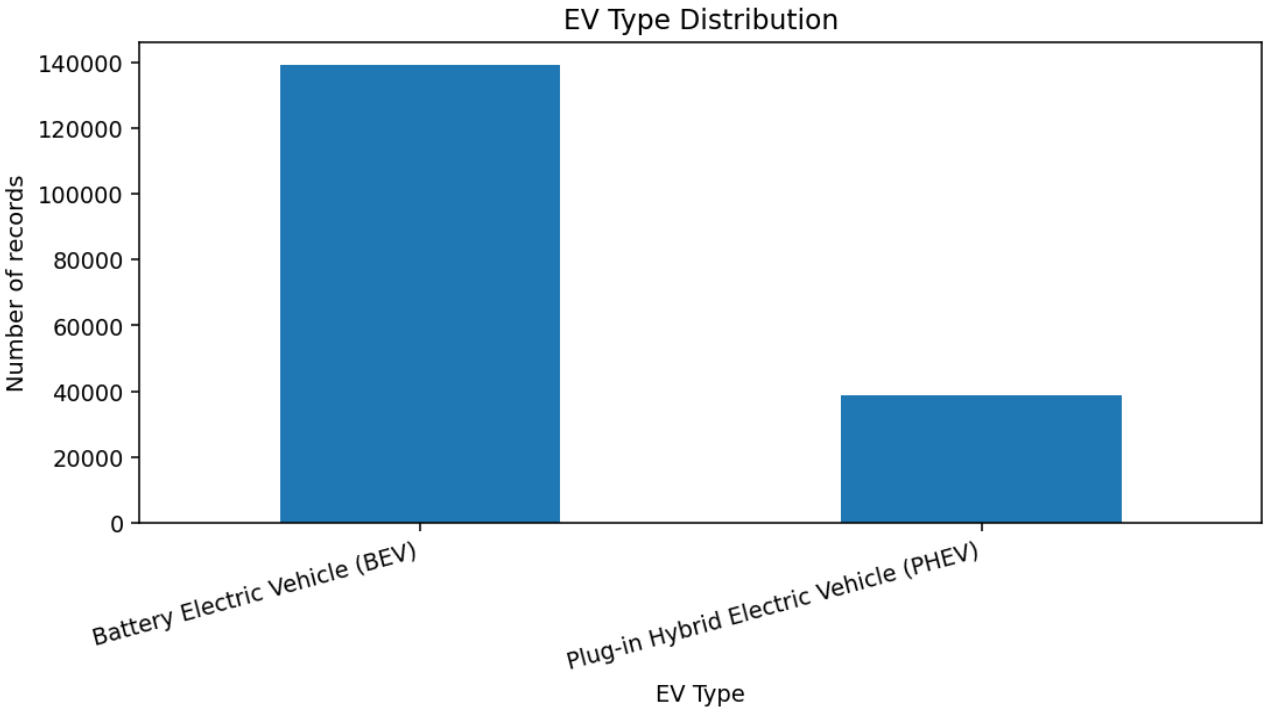
Geographic distribution and make popularity



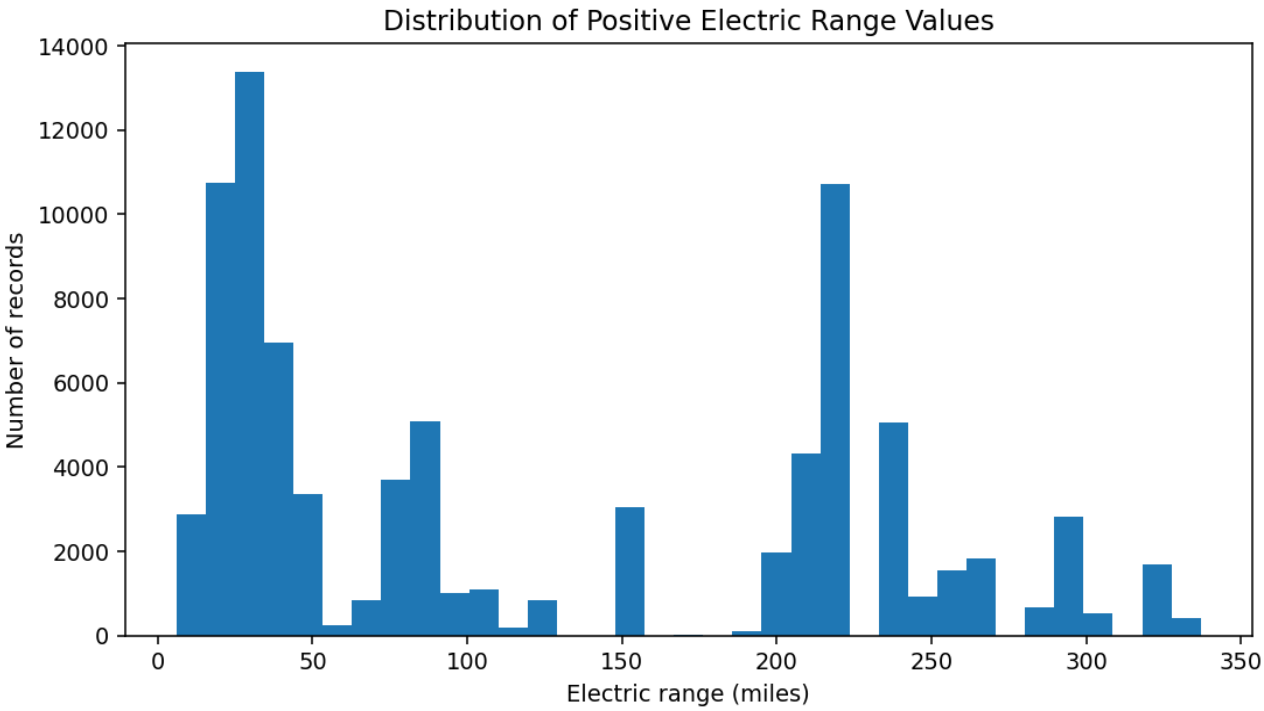

Top 10 EV Makes
TESLA
NISSAN
CHEVROLET
FORD
BMW
KIA
TOYOTA
VOLKSWAGEN
JEEP
HYUNDAI
Make
0
10000
20000
30000
40000
50000
60000
70000
80000
Number of records

Popular models and EV type



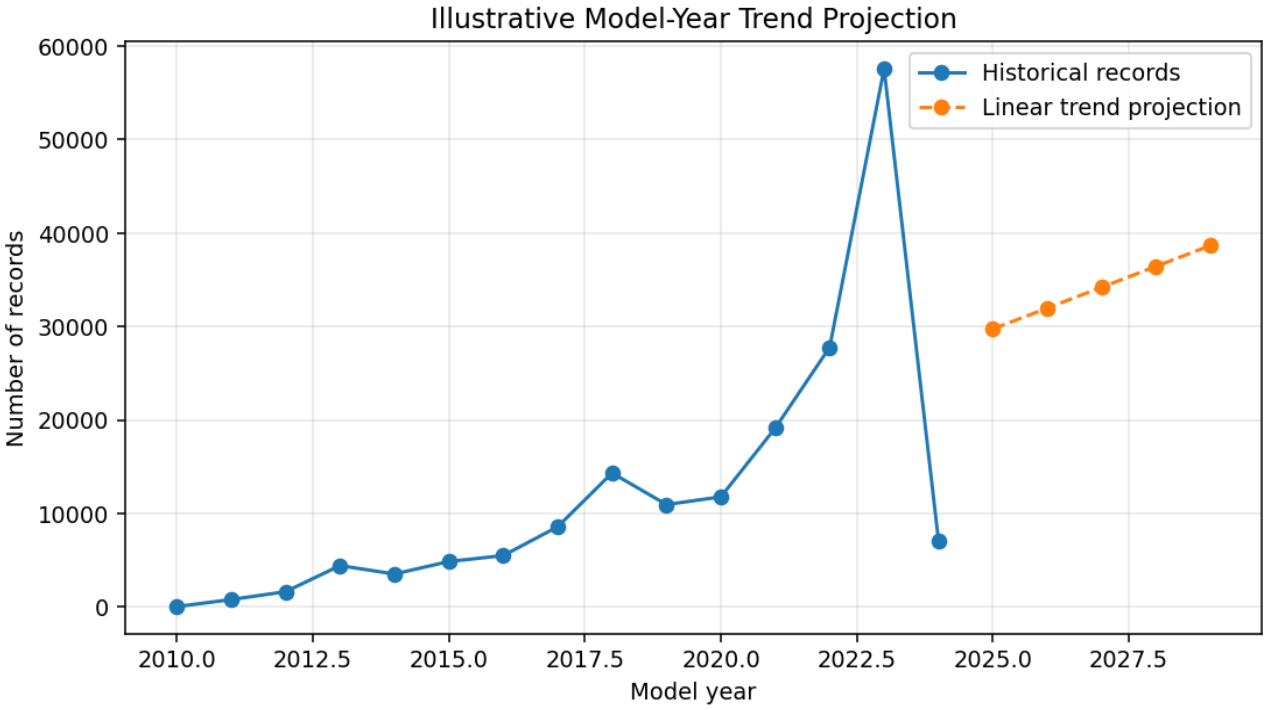


Electric range distribution



6. Forecasting and Validation

A simple linear regression was fitted to the annual count of records by model year and evaluated on the latest held-out observations. The holdout metrics were MAE = 20,903.5, RMSE = 24,507.0, and R^2 = -0.397. The negative R^2 indicates that the simple linear model performed poorly on the holdout period and should not be relied upon for precise forecasting. The dataset is a snapshot, and model year is not registration year; future projections are therefore illustrative only.



7. Geospatial Clustering

Vehicle Location coordinates were parsed from the POINT(longitude latitude) text. Usable coordinates were available for 177,857 rows. K-means clustering with five clusters was used on a reproducible sample of up to 10,000 coordinates as an exploratory hotspot demonstration. Cluster labels are algorithmic groups, not administrative boundaries or proof of causal adoption factors.

8. Recommendations

1. Use county and city counts to prioritise deeper infrastructure planning, then normalise by population and charging availability before comparing adoption rates.
2. Examine make/model and EV-type shares to understand the mix represented in the registration snapshot.
3. Treat zero range and MSRP values carefully; distinguish unavailable values from true zeroes before pricing or range conclusions.
4. Do not interpret correlation as causation. Use vehicle-level factors and a documented model before making price/range claims.
5. Improve forecasting with a time-stamped registration series, actual registration dates, population denominators and model validation against a chronological holdout.

9. Limitations and Assumptions

• The workbook is a snapshot with model year, not a time series of registration dates. It cannot by itself establish year-over-year registrations or total market revenue.
• The file has a strong Washington State focus, as stated in the brief; findings should not be generalised to all US EVs.
• Base MSRP values can be zero and may not reflect current transaction prices.
• County/city counts are not per-capita adoption rates because population data were not supplied.
• Forecast validation is weak (negative R^2), so projections are illustrative only.

10. Conclusion

The supplied EV population data show a concentrated pattern of records by geography, manufacturer and model, and offer useful descriptive evidence about range and EV types. The analysis provides reproducible cleaning, descriptive statistics, visualisations, coordinate parsing, exploratory clustering and a transparent forecast-validation workflow. More reliable market sizing requires dated registration flows, population denominators, and sales/revenue data.

Appendix: Dataset fields

The notebook includes the full 17-column source schema, cleaning code, missing-value audit, annual model-year counts, county/city summaries, make/model/type distributions, range/MSRP analysis, geospatial clustering, forecast evaluation, and recommendations.